Discrete Time Fourier Transform (DTFT)

$\{x(n)\}$ → DT Signal

The DTFT of $\{x(n)\}$ is defined as

DTFT

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$

$-\infty < \omega < \infty$

ω → real variable (frequency)

$X(\omega)$ ⇒ spectrum of $\{x(n)\}$

$X(\omega) : \mathbb{R} \longrightarrow \mathbb{C}$

$\omega \qquad X(\omega)$

$|X(\omega)|$ ⇒ Magnitude Spectrum

$\angle X(\omega)$ ⇒ phase spectrum

$X(\omega)$ ⇒ reveals about frequency contents in the signal $\{x(n)\}$

Inverse DTFT

$$x(n) = \frac{1}{2\pi} \int_{\omega=-\pi}^{\pi} X(\omega)\, e^{j\omega n}\, d\omega$$

$x(n)$: DT Signal

$\int$: Summation

$X(\omega)$: Weight

$e^{j\omega n}$: DT complex exponential with freq. ω (radians/sample)

Properties of DTFT

(1) $X(\omega)$ is a periodic function of ω with period 2π

$$X(\omega) = X(\omega + 2\pi) \quad \text{for any } \omega \in \mathbb{R}$$

Proof:

$$X(\omega + 2\pi) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j(\omega+2\pi)n}$$

$$= \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}\, \underbrace{e^{-j2\pi n}}_{1}$$

$$= X(\omega)$$

(2) Relation between DTFT & Z-Transform

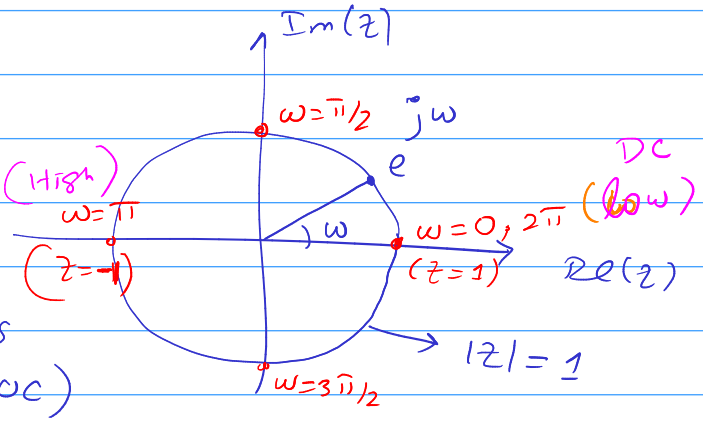
$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n} \quad ; \text{ ROC } R$$

Let us assume unit circle $|z| = 1$ is in ROC

We can find

$X(z)$ @ $z = e^{jw}$

(since $|z| = 1$ is in ROC)



$$X(z)\Big|_{@\, z=e^{jw}} = \sum_{n=-\infty}^{\infty} x(n)\, e^{-jwn}$$

$$X(e^{jw}) = X(w)$$

$X(e^{jw})$ ⇓ Value of z transform at $z = e^{jw}$

$X(w)$ ⇓ Spectrum @ frequency w

3) Existence of DTFT

Sufficient conditions

- If $\{x(n)\}$ is absolutely summable then DTFT $X(\omega)$ exists

- If $\{x(n)\}$ has finite energy

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 < \infty$$

then DTFT $x(\omega)$ exists



Examples of DTFT

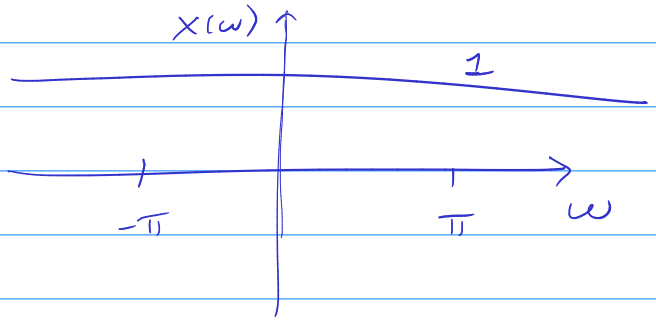
① $\{x(n)\} = \{a, b, c, d, e\}$ (n = 0 at b)

$$X(\omega) = a e^{+j\omega} + b e^{j\omega(0)} + c e^{-j\omega} + d e^{-j2\omega} + e^{-j3\omega}$$

$-\infty < \omega < \infty$

② $\{x(n)\} = \{\delta(n)\}$

$X(\omega) = 1$ for all ω



③ $x(n) = a^n u(n)$ $\quad X(z) = \dfrac{1}{1 - az^{-1}}$; $|z| > |a|$

$|a| < 1$

ROC includes unit circle

Spectrum at frequency $\omega \Rightarrow X(\omega) = X(z)\Big|_{z = e^{j\omega}}$

$$= \frac{1}{1 - a e^{-j\omega}}$$

$$X(\omega) = \frac{1}{1 - a\cos\omega + j\sin\omega}$$

$$|X(\omega)| = \frac{1}{\sqrt{1+a^2-2a\cos\omega}}$$

$$\angle X(\omega) = -\tan^{-1}\left(\frac{\sin\omega}{1-a\cos\omega}\right)$$